

Virtual Data Science with Python Apprentice Intern

Week 1 Report

Topic:

Data Acquisition, Cleaning and Exploratory Data Analysis using Python

Submitted By: Khalik Ansari

Internship: NSDC Virtual Data Science with Python Apprentice Intern

Week: Week 1

Submission Date: 28 July 2026

1. Objective

The objective of this task is to acquire a publicly available dataset, clean and preprocess the data using Python, perform exploratory data analysis (EDA), create meaningful visualizations, and identify useful insights from the dataset.

2. Dataset Information

- **Dataset Name:** Titanic_Dataset
- **Source:** Kaggle
- **Total Rows:** 891
- **Total Columns:** 12
- **File Format:** CSV

3. Libraries Used

The following Python libraries were used during this project:

- **Pandas** : For loading, cleaning, and analyzing the dataset.
- **Matplotlib** : For creating visualizations and charts.
- **Seaborn** : For creating statistical and attractive visualizations.

4. Data Loading

The Titanic dataset was loaded into Python using the Pandas library. The `read_csv()` function was used to read the CSV file and store it in a DataFrame named `df`. This DataFrame was then used for data exploration, cleaning, and analysis.

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
df = pd.read_csv("Titanic-Dataset.csv")
```

```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

The `df.head()` function displays the first five rows of the dataset. It helps verify that the dataset has been loaded correctly and provides an overview of the available columns and sample records.

4. Data Exploration

4.1 Dataset Shape:

Code: `df.shape`

```
df.shape
(891, 12)
```

The `df.shape` function was used to determine the size of the dataset. The Titanic dataset contains **891 rows** and **12 columns**, indicating the number of observations and features available for analysis.

4.2 Dataset Information:

Code: `df.info()`

```
df.info()

<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype  
---  --
 0   PassengerId    891 non-null    int64  
 1   Survived       891 non-null    int64  
 2   Pclass        891 non-null    int64  
 3   Name           891 non-null    str     
 4   Sex            891 non-null    str     
 5   Age           714 non-null    float64 
 6   SibSp          891 non-null    int64  
 7   Parch         891 non-null    int64  
 8   Ticket         891 non-null    str     
 9   Fare           891 non-null    float64 
10   Cabin         204 non-null    str     
11   Embarked      889 non-null    str     
dtypes: float64(2), int64(5), str(5)
memory usage: 118.9 KB
```

The `df.info()` function provides detailed information about the dataset, including the total number of columns, data types, and missing values. It showed that the **Age**, **Cabin**, and **Embarked** columns contained missing values.

4.3 Statistical Summary:

Code: `df.describe()`

```
df.describe()


```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

The `df.describe()` function provides summary statistics such as count, mean, standard deviation, minimum, maximum, and quartile values for numerical columns. It helped understand the distribution of numerical features like Age and Fare.

5. Data Cleaning

Data cleaning is an essential step in data science. It improves the quality of the dataset by handling missing values, removing unnecessary data, and preparing the data for analysis

5.1 Checking Missing Values:

Code: `df.isnull().sum()`

```
df.isnull().sum()
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age           177
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin         687
Embarked       2
dtype: int64
```

The `isnull().sum()` function was used to identify missing values in each column. The dataset contained missing values in the **Age**, **Cabin**, and **Embarked** columns.

5.2 Checking Duplicate Records

Code: `df.duplicated().sum()`

```
df.duplicated().sum()
np.int64(0)
```

The dataset was checked for duplicate records using the `duplicated().sum()` function. No duplicate rows were found in the dataset.

5.3 Handling Missing Values in Age

Code: `df["Age"] = df["Age"].fillna(df["Age"].median())`

```
df["Age"] = df["Age"].fillna(df["Age"].median())
✓ 0.0s
```

Missing values in the **Age** column were replaced with the **median** value because the median is less affected by extreme values and provides a reliable estimate.

5.4 Handling Missing Values in Embarked

Code: `df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])`

```
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
✓ 0.0s Python

df["Embarked"].isnull().sum()
✓ 0.0s Python

np.Int64(0)
```

Missing values in the **Embarked** column were replaced with the **mode**, which represents the most frequently occurring category.

5.5 Removing the Cabin Column

Code: `df = df.drop(columns=["Cabin"])`

```
df = df.drop(columns=["Cabin"])
✓ 0.0s Python

df.columns
✓ 0.0s Python

Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
      'Parch', 'Ticket', 'Fare', 'Embarked'],
      dtype='str')
```

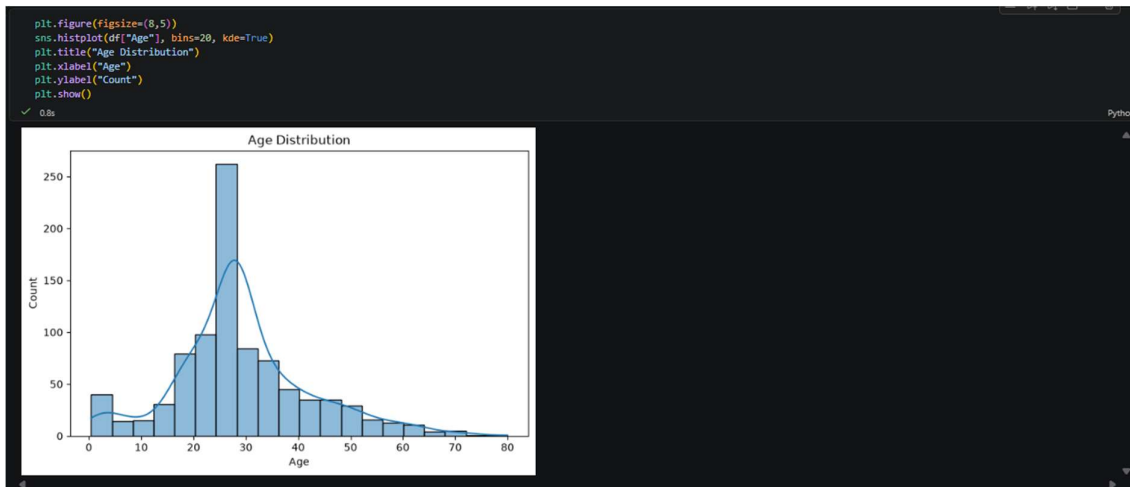
The **Cabin** column contained a large number of missing values (687 out of 891 records). Therefore, it was removed from the dataset to improve data quality and simplify further analysis.

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the distribution of the data, identify trends, and discover relationships among different variables using visualizations.

6.1 Visualization 1: Age Distribution (Histogram)

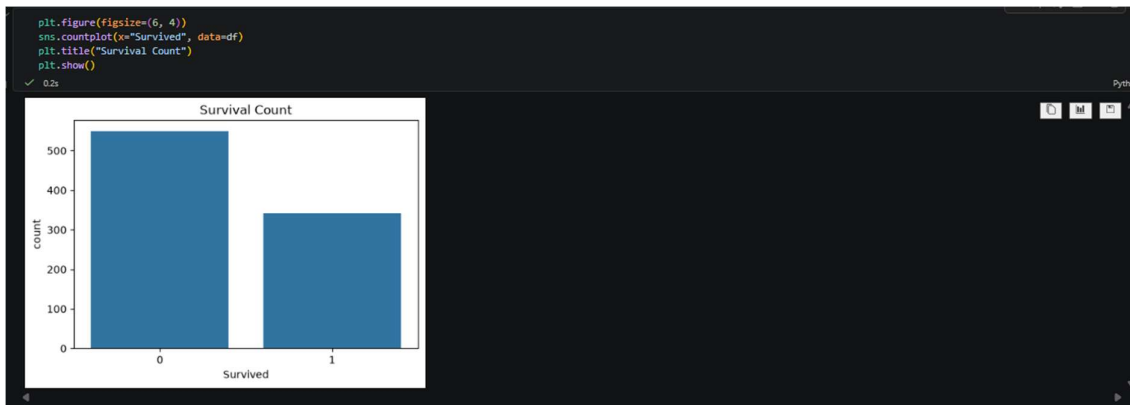
Code: `plt.figure(figsize=(8,5))`
`sns.histplot(df["Age"], bins=20, kde=True)`
`plt.title("Age Distribution")`
`plt.xlabel("Age")`
`plt.ylabel("Count")`
`plt.show()`



The histogram shows the distribution of passengers by age. Most passengers were between **20 and 35 years** old, while only a few passengers were above **60 years** of age.

6.2 Visualization 2: Survival Count (Bar Chart)

Code: `plt.figure(figsize=(6,4))`
`sns.countplot(x="Survived", data=df)`
`plt.title("Survival Count")`
`plt.show()`



The survival count chart shows that the number of passengers who **did not survive** was higher than the number of passengers who survived the Titanic disaster.

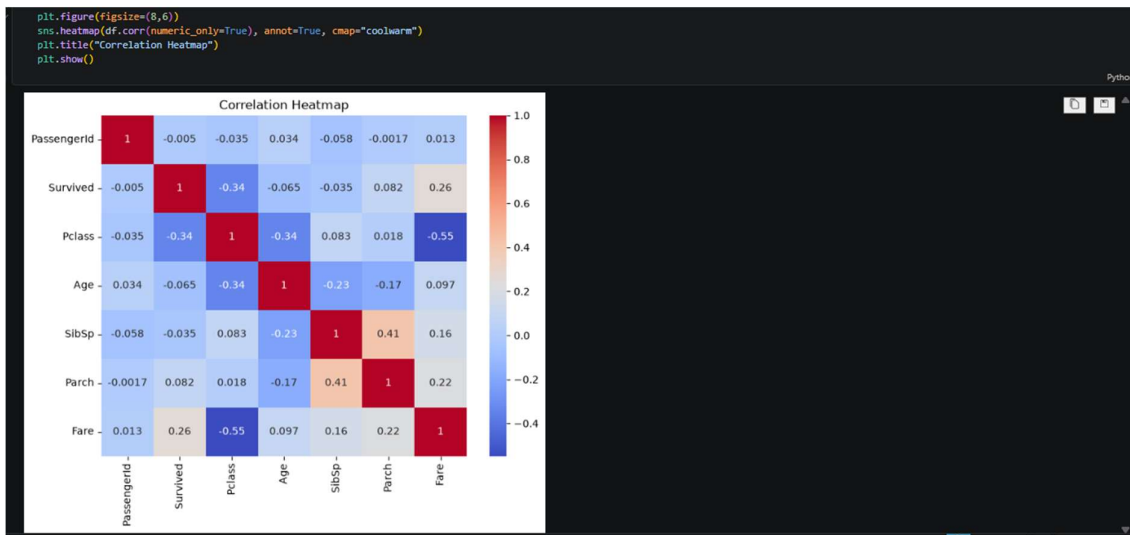
6.3 Visualization 3: Correlation Heatmap

Code: plt.figure(figsize=(8,6))

sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")

plt.title("Correlation Heatmap")

plt.show()



The correlation heatmap illustrates the relationship between numerical features. A negative correlation was observed between **Passenger Class (Pclass)** and **Fare**, while **SibSp** and **Parch** showed a moderate positive correlation.

7. Key Insights

The exploratory data analysis of the Titanic dataset revealed the following key insights:

- 1. Most passengers were between 20 and 35 years of age**, indicating that the majority of travelers were young adults.
- 2. The number of passengers who did not survive was greater than the number who survived**, showing that the survival rate was relatively low.
- 3. The Age column contained 177 missing values**, which were successfully replaced using the **median** to preserve the data distribution.
- 4. The Embarked column had 2 missing values**, and these were filled using the **mode**, as it represents the most common category.
- 5. The Cabin column contained 687 missing values**, making it unsuitable for analysis. Therefore, it was removed from the dataset.
- 6. No duplicate records were found** in the dataset, indicating that the data was already free from duplicate entries.
- 7. The correlation heatmap showed meaningful relationships between numerical features**, such as a negative correlation between **Passenger Class (Pclass)** and **Fare**, and a moderate positive correlation between **SibSp** and **Parch**.

8. Conclusion

The Titanic dataset was successfully loaded, explored, cleaned, and analyzed using Python. Missing values were handled appropriately, unnecessary data was removed, and the dataset was prepared for analysis. Exploratory Data Analysis (EDA) was performed using various visualizations to understand data patterns and relationships. The analysis provided valuable insights into passenger demographics and survival trends. This project improved practical skills in data preprocessing, data visualization, and data analysis using Python libraries such as Pandas, Matplotlib, and Seaborn.

9. Future Scope

This project can be further enhanced in the following ways:

- Build a machine learning model to predict passenger survival.
- Perform feature engineering to improve prediction accuracy.
- Explore additional visualizations for deeper insights.
- Analyze the impact of passenger class, gender, and fare on survival.
- Develop an interactive dashboard using Power BI, Tableau, or Streamlit to present the analysis visually.